

Autoencoders and Variational Autoencoders on MNIST

Kiran Fatima (24L-8003)
Email: 1248003@lhr.nu.edu.pk

Abstract—This report presents the implementation and analysis of a Bottleneck AE and VAE on MNIST. The AE is evaluated across bottleneck dimensions $\{100, 50, 30, 20\}$ with MSE error reported for each. The VAE is implemented with a 2D latent space using the reparameterization trick and extended to 10D, with comparative analysis across latent visualization, standard normal sampling, and uniform sampling. All inputs are normalized to $[0, 1]$ via `ToTensor()`, and all models are trained for 20 epochs using Adam (lr= 0.001), batch size 128.

I. AE IMPLEMENTATION

A. Architecture, Code Structure and Training

The implementation is organized into separate notebook cells for model definition, loss computation, training loop, and visualization utilities, satisfying the well-structured code requirement. A symmetric three-layer AE (784→512→256→ d →256→512→784) with ReLU activations in hidden layers and a Sigmoid output was trained for 20 epochs. The Sigmoid output constrains pixel values to $[0, 1]$, consistent with the normalized MNIST inputs. MSE loss is used rather than BCE because the AE treats pixel values as continuous intensities rather than Bernoulli probabilities. Training used the Adam optimizer (lr= 0.001) with a batch size of 128.

B. Reconstruction Quality and Error



Fig. 1. AE reconstructions for all bottleneck dimensions. Each panel shows original digits (top row) vs. reconstructed digits (bottom row) for the labeled bottleneck dimension.

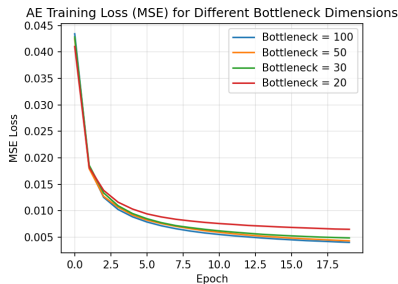


Fig. 2. AE training loss (MSE) curves for all four bottleneck dimensions across 20 epochs.

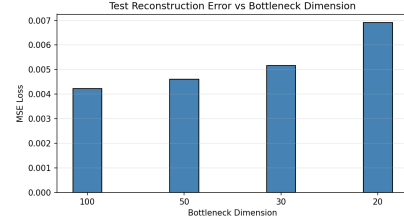


Fig. 3. Test reconstruction error (MSE) vs. bottleneck dimension.

TABLE I
TEST MSE RECONSTRUCTION ERROR PER BOTTLENECK DIM

Bottleneck Dim	Test MSE
100	0.004221
50	0.004611
30	0.005163
20	0.006909

Reconstruction error increases monotonically with compression. The dim= 100→50 gap is small (0.004221 vs. 0.004611), indicating diminishing returns beyond 50 dimensions. The larger dim= 30→20 jump (0.005163→0.006909) marks the compression threshold below which digit structure is irreversibly discarded complex digits such as 8 and 5 exhibit visible blurring at dim= 20, while simpler digits like 0 and 1 remain clean across all settings.

II. VAE IMPLEMENTATION

A. Architecture, Reparameterization, and Training

The encoder (784→512→256) produces two heads, $\mu \in R^d$ and $\log \sigma^2 \in R^d$, from a shared hidden representation. The latent variable is drawn via the reparameterization trick:

$$\mathbf{z} = \mu + \epsilon \odot e^{\frac{1}{2} \log \sigma^2}, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (1)$$

keeping the stochastic node outside the computational graph so gradients flow through μ and $\log \sigma^2$. The decoder (latent→256→512→784, Sigmoid) outputs per-pixel Bernoulli probabilities, motivating the VAE loss:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} - \frac{1}{2} \sum_j (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2) \quad (2)$$

Both models converge by epoch 15–18 and are saved as `vae_2d.pth` and `vae_10d.pth`.



Fig. 4. VAE training loss comparison for 2D and 10D latent spaces. Metrics: Total, BCE, KL.

TABLE II
FINAL EPOCH LOSS: 2D VS. 10D VAE

Model	Total	Recon (BCE)	KL
2D VAE	139.59	132.86	6.73
10D VAE	100.66	81.78	18.88

B. Latent Space Visualization

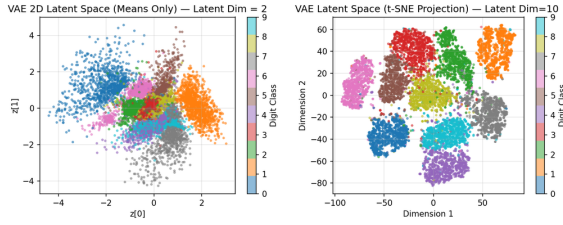


Fig. 5. Encoder means plotted in latent space. Left: 2D direct. Right: 10D via t-SNE.

In the **2D** latent space, the digits form recognizable clusters rather than being scattered randomly. Each class occupies a broadly distinguishable region, but visually similar pairs such as $\{4, 9\}$ and $\{3, 8\}$ overlap due to shared stroke patterns. The KL constraint compresses all ten classes into a compact $\mathcal{N}(0, \mathbf{I})$ region, making complete separation impossible with only two dimensions. In **10D**, the t-SNE projection reveals well-defined, minimally overlapping clusters for all digit classes, indicating substantially better class separation than in the 2D case. The additional dimensions give the encoder sufficient capacity to push each class into its own distinct region of the latent manifold, a substantial improvement that justifies the increased KL cost ($6.73 \rightarrow 18.88$) as a worthwhile trade-off for more discriminative latent representations.

C. Generation: Normal and Uniform Sampling

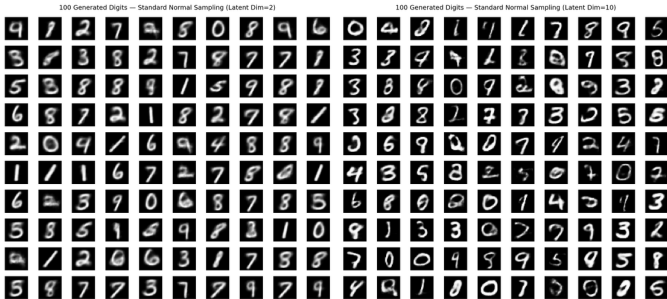


Fig. 6. 10×10 grids from $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Left: 2D. Right: 10D.

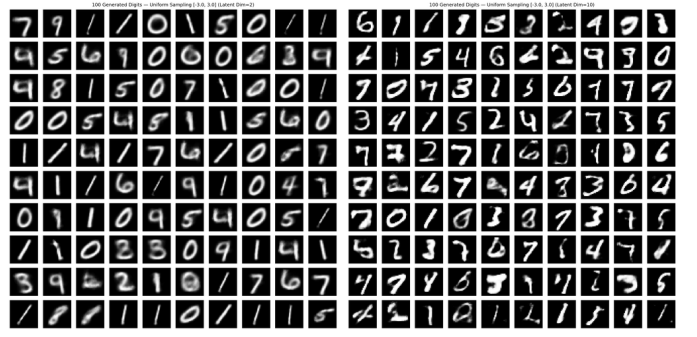


Fig. 7. 10×10 grids from uniform $[-3, 3]^d$. Left: 2D (mild holes). Right: 10D (severe holes).

Standard normal sampling produces smooth, continuous digit transitions in both dimensionalities. In **2D**, adjacent grid cells morph gradually between digit styles (e.g., $6 \rightarrow 0$, $7 \rightarrow 1$) with no blank outputs, since $\mathcal{N}(\mathbf{0}, \mathbf{I})$ concentrates samples precisely in the trained region. In **10D**, transitions are smoother and sharper — the richer latent space encodes finer stroke details, producing more realistic handwriting variation while preserving the continuous interpolation property across all ten digit classes.

Uniform sampling over $[-3, 3]^d$ exposes latent holes in both cases, but with starkly different severity. In **2D**, holes are mild: most uniform samples land near the trained region since $[-3, 3]^2$ is small, yielding recognizable digits with only edge distortions. In **10D**, holes are far more severe $[-3, 3]^{10}$ has volume $6^{10} \approx 6 \times 10^7$, while the KL prior confines the learned manifold to a tiny neighborhood near the origin. Nearly all uniform samples therefore fall outside any region visited during training, and the decoder produces meaningless noise. Holes in both cases represent latent coordinates the encoder never visited — the decoder has no valid mapping there. This curse of dimensionality confirms that $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is the only correct sampling strategy for VAEs regardless of latent dimensionality.

TABLE III
2D VS. 10D VAE SUMMARY

Aspect	2D	10D
BCE Loss	132.86	81.78 (38%↓)
Cluster separation	Partial	Clear (t-SNE)
Normal sampling	Smooth	Smoother, sharper
Uniform holes	Mild	Severe (curse of dim.)

III. CONCLUSIONS

AE reconstruction degrades gracefully with compression, steepest below $\text{dim} = 30$. VAE latent clusters improve dramatically from 2D to 10D, accompanied by sharper generation quality under standard normal sampling. Uniform sampling confirms latent holes are mild in 2D but severe in 10D due to the exponentially larger untrained volume, making $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ the appropriate generation strategy for VAEs.

Declaration: I completed this assignment independently and did not take help from any other person.